Qualcomm Developer Project
Stable Diffusion 2.1 Demo on QCS9075 LE

Attribution:	
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Project Title*	Stable Diffusion 2.1 Depolyment on QCS9075/QCS9100 LE
Project Description* High-Level Project Description (75 words or fewer)	This project implements a command-line application in Python for text-to-image generation based on Stable Diffusion2.1. It enables converting textual descriptions into corresponding images, and leverages Qualcomm QNN to accelerate the inference process, making the image generation more efficient on compatible hardware platforms.
Images Upload up to 5 images of your project **Please submit/send the original JPEG/PNG files for all images included in the document**	 Generated by the prompt: "A kitten is practicing martial arts"
Objective -What inspired you to create this project? -What is your desired outcome?	The inspiration for this project comes from the desire to make advanced text-to-image generation technology more accessible and efficient on edge devices. While Stable Diffusion 2.1 has revolutionized AI-generated imagery with its high-quality output, its computational intensity often restricts it to powerful GPUs or cloud environments, creating barriers for users who need quick, local generation without relying on network connectivity. Additionally, most existing Stable Diffusion applications are GUI-based, which can be cumbersome for developers or power users who prefer lightweight, scriptable tools for batch processing or integration into other workflows. This gap—between the potential of Stable Diffusion 2.1 and its accessibility on edge hardware—motivated me to build a Python command-line interface (CLI) app. Leveraging Qualcomm QNN was a natural choice to address the efficiency challenge, as it optimizes neural network inference on Qualcomm hardware, enabling faster, more power-efficient execution of large models like Stable Diffusion 2.1 locally. The combination of a CLI’s simplicity and QNN’s acceleration perfectly aligns with the need for a practical, low-barrier tool that brings AI-generated imagery closer to everyday edge computing scenarios. My desired outcome is to deliver a lightweight, high-performance CLI tool that makes Stable Diffusion 2.1’s text-to-image capability easily accessible on

	Qualcomm-powered edge devices. Specifically, I aim to validate that the app can generate high-fidelity images from text prompts within seconds (under 5 seconds per image for standard resolutions) using QNN acceleration, outperforming unoptimized local implementations. Beyond technical validation, I hope this tool serves as a flexible prototype for developers—enabling them to integrate text-to-image generation into automation scripts, batch processing pipelines, or custom edge AI applications. Ultimately, this project seeks to democratize access to advanced generative AI on edge hardware, showcasing how QNN can unlock the potential of large diffusion models in resource-constrained environments and inspiring more practical, edge-native generative AI tools.
Operating System* **（Android，Linux，Windows 10IoT Core）**	☐ Android ☑ Linux ☐ ThreadX RTOS ☐ Windows 10IoT Core ☐ UbuntuCore ☐ WindowsCore
Cloud Services/Platform	☐ Sierra Wireless AirVantage ☐ AT&T M2x ☐ IBM Bluemix ☐ IBM Watson IoT ☐ Gizwits Cloud Platform ☐ Microsoft Azure IoT
Skill Level Required	☐ Advanced ☑ Beginner ☐ Intermediate
Areas of Focus	☐ 3D Printing & Modeling ☐ Alexa Voice Service √ Artificial Intelligence ☐ Bluetooth ☐ Computer Vision ☐ Digital Signage ☐ Education √ Embedded ☐ Gaming ☐ Healthcare ☐ IoT ☐ Robotics ☐ Security ☐ Sensors ☐ Smart Cities ☐ Smart Home ☐ Toys
Materials Required / Parts List / Tools	Part Name: TurboX C9100 Link to purchase:
Source Code / Source Examples / Application Executable *Link to open source / shareable code repository*	Description: Source Code Link: https://github.com/ThunderSoft-XA/Stable-diffusion-2.1-deployment-on-QCS9075-LE/tree/master
Additional Resources *List related links or resources such as websites, videos, presentations, or other materials*	Resource Title: Test encoder, unet, vae model files Link or File Name (and provide file): https://huggingface.co/qualcomm/Stable-Diffusion-v2.1/tree/main/precompiled/qualcomm-qcs9075-proxy Stable-Diffusion-v2.1_text_encoder_w8a16.bin Stable-Diffusion-v2.1_unet_w8a16.bin Stable-Diffusion-v2.1_vae_w8a16.bin
Build / Assembly Instructions	1. Follow the guide in the document under https://github.com/ThunderSoft-XA/Stable-diffusion-2.1-deployment-on-QCS9075-LE/tree/master to install the

	right QAIRT SDK QCS9075 or QCS9100 LE device. 2. Clone the project on the device
Usage Instructions	Run the command line like : python3 sd21_qnn_linux.py --prompt "A kitten is practicing martial arts" --steps 20 --seed 1 --guidance 7.5 --output sd21_qnn.png

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